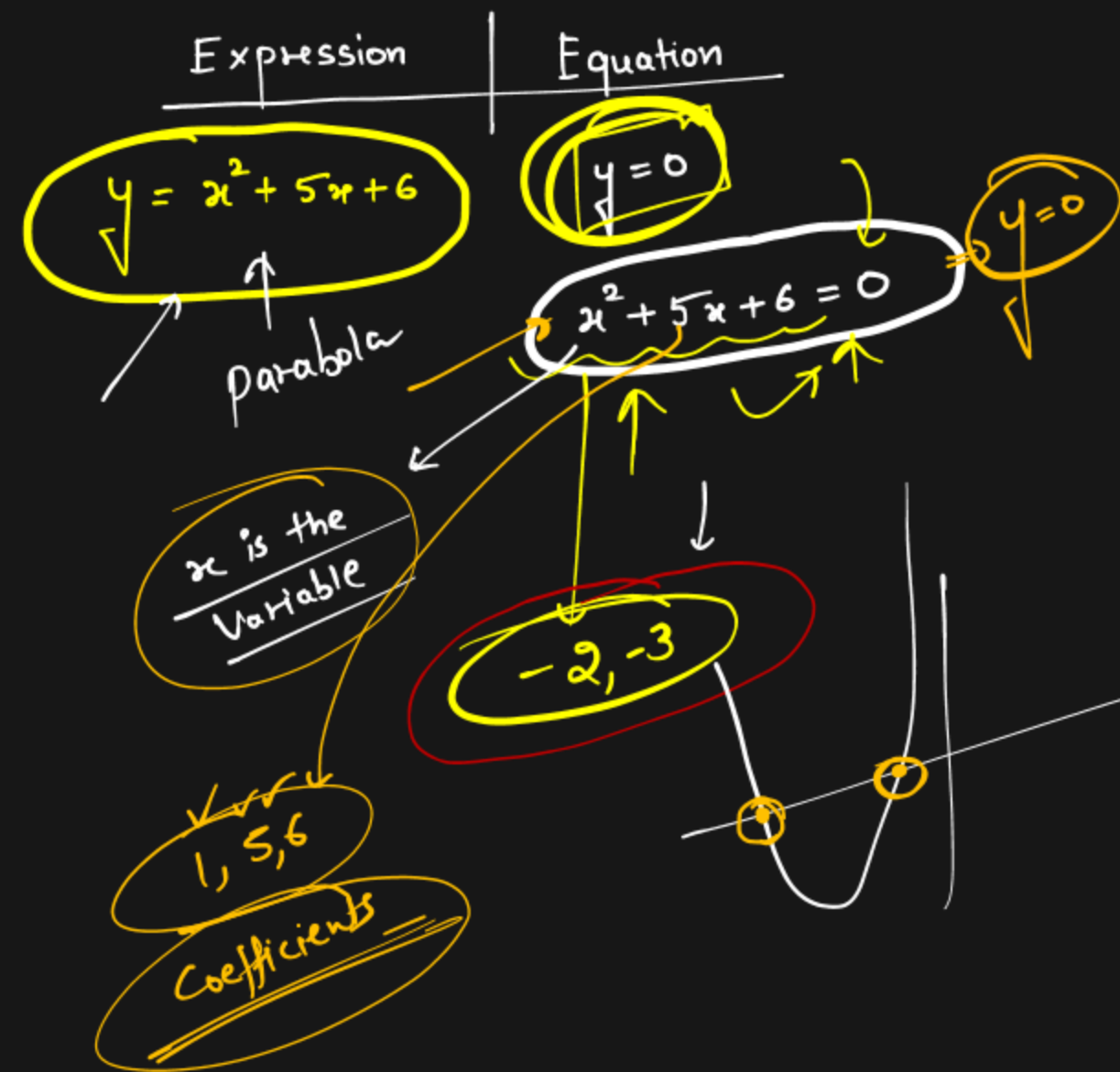




$$ax^2 + bx + c$$

Expression

$$a = 1$$

$$b = 5$$

$$c = 6$$

$$x^2 + 5x + 6$$

1.) Linear Equations $\rightarrow P, M$

2.) Quadratic Equations $\rightarrow P, M$

3.) Cubic Equations $\rightarrow M$

$\rightarrow$ Variable की Max. Power = 1

$\rightarrow$ Variable की Max Power = 2

$\rightarrow$ Variable की Max Power = 3

Degree

$$4x + 5 = 0 \quad 2x + 3y = 0$$

$$2y + 9 = 0$$

$$4x + 3y + 5z = 0$$

$$x^2 + 7x + 12 = 0$$

$$y^2 + 13x + 30 = 0$$

$$p^3 + 3p^2 + 7p + 8 = 0$$

Linear Equations

Linear Equations with one variable

$$4x + 2 = 0$$

$$4x = -2$$

$$x = -\frac{2}{4}$$

$$x = -\frac{1}{2}$$

$$14x - 28 = 0$$

$$13a + 6 = 0$$

$$14x = 28$$

$$x = 2$$

$$13a = -6$$

$$a = -\frac{6}{13}$$

Linear Equations with two variables

$$4x + 2y = 0$$

$$7x + 9y = 8$$

$$14x + 3y = 40$$

Linear equations with three variables

$$3x + 2y + 9z = 15$$

$$4a + 7b + 9c = 36$$

$$8p + 9q + 7r = 25$$

Solving Linear Equations :

Linear Equations Solution Rule

$$\text{No. of variables} = \text{No. of Equations}$$

- | | | |
|-----------------|---|---------------------------------------|
| One Variable | → | One equation is required |
| Two Variables | → | Two equations ^{are} required |
| Three Variables | → | Three equations are required |

$$ax + by = c$$

$$px + qy = r$$

$$\frac{a}{p} = \frac{b}{q}$$

parallel lines

They will not intersect

$x, y \Rightarrow \underline{\underline{CND}}$

we have to
find out

the values of x and y
that satisfies both the equations
[that is point of Intersection]

Quadratic Equations

Variable की
Max. Power = 2

Basic form $\Rightarrow ax^2 + bx + c = 0$

$\rightarrow \alpha, \beta$

$$ax^2 + bx + c = 0$$

- * x is the variable
- * a, b, c are called coefficients

Degree = 2

Solutions = 2

Roots = 2

E.g.

$$x^2 + 6x + 8 = 0$$

$$a=1, b=6, c=8$$

Sum and Product of Roots :

$$ax^2 + bx + c = 0 \rightarrow (\alpha, \beta)$$

$$\text{Sum} = \alpha + \beta = -\frac{b}{a} \quad \checkmark$$

$$\text{Product} = \alpha \cdot \beta = +\frac{c}{a} \quad \checkmark$$

$$-x^2 + 5x + 6 = 0$$

$$[-3, -2]$$

$$\text{Sum} = -5$$

$$\text{Product} = +6$$

$$\frac{-5}{1} \Rightarrow -5 \quad \checkmark$$

$$\frac{+6}{1} \Rightarrow 6 \quad \checkmark$$

If we are given with Sum of Roots and Product of Roots, We can form equation also.

$$x^2 - [\text{Sum}]x + [\text{Prod}] = 0$$

$$x^2 - \left[-\frac{b}{a}\right]x + \left[\frac{c}{a}\right] = 0$$

$$x^2 + \frac{bx}{a} + \frac{c}{a} = 0$$

$$\frac{ax^2 + bx + c}{a} = 0$$

$$ax^2 + bx + c = 0$$

Root of an equation always satisfies the equation :

$x^2 + 5x + 6 = 0$ $\rightarrow [-3, -2]$

$$x = (-3)$$

$$(-3)^2 + 5[-3] + 6 = 0$$

$$9 - 15 + 6 = 0$$

$$15 - 15 = 0$$

$$0 = 0$$

$$x = (-2)$$

$$(-2)^2 + 5(-2) + 6 = 0$$

$$4 - 10 + 6 = 0$$

$$10 - 10 = 0$$

$$0 = 0$$

* Discriminant = $b^2 - 4ac$

$$ax^2 + bx + c = 0$$

✓✓
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

✓✓
$$x = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

Nature of Roots :-

- (i) $D = b^2 - 4ac > 0$ $\rightarrow$ Roots are Real, Unequal
- (ii) $D = b^2 - 4ac = 0$ $\rightarrow$ Roots are Real, Equal
- (iii) $D = b^2 - 4ac < 0$ $\rightarrow$ Roots are Imaginary

$b^2 - 4ac$ is a non perfect sq.

Roots are Irrational

$b^2 - 4ac$ is a perfect sq.

Roots are Rational

Cubic Equations

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$$ax^3 + bx^2 + cx + d = 0$$

$$\boxed{m, n, o}$$

$$m + n + o = -\frac{b}{a}$$

$$mn + no + om = \frac{c}{a}$$

$$m \cdot n \cdot o = -\frac{d}{a}$$

Sum and Prod. of Roots :

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	Sum	Product
1.) $x^2 + 3x + 8 = 0$	$-\frac{3}{1} \Rightarrow -3$	$\frac{8}{1} \Rightarrow 8$
2.) $9y^2 + 7y + 14 = 0$	$-\frac{7}{9}$	$\frac{14}{9} \Rightarrow 14/9$
3.) $2y^2 - 8y + 17 = 0$	$-\frac{(-8)}{2} = (+4)$	$\frac{17}{2}$
4.) $3x^2 - 14x - 20 = 0$	$-\frac{(-14)}{3} \Rightarrow \frac{+14}{3}$	$\frac{-20}{3}$
5.) $5x^2 - 4x - 18 = 0$	$-\left(\frac{-4}{5}\right) \Rightarrow \frac{4}{5}$	$\frac{-18}{5}$

Create Quadratic Equation using Sum of Roots and Product of Roots $\Rightarrow$

$$x^2 - [\text{Sum}]x + [\text{Product}] = 0$$

	Sum	Product	
①	10	30	$x^2 - 10x + 30 = 0$
②	7	12	$x^2 - 7x + 12 = 0$
③	14.5	-30	$x^2 - (14.5)x + (-30) = 0 \Rightarrow 2x^2 - 29x - 60 = 0$
④	$\frac{4}{3}$	$\frac{15}{2}$	$x^2 - \frac{4}{3}x + \frac{15}{2} = 0 \Rightarrow 6x^2 - 8x + 45 = 0$
⑤	-17	70	$x^2 - (-17)x + 70 = 0$ $x^2 + 17x + 70 = 0$ ✓

Find the Nature of Roots

Real, Unequal, Rational

Real, Unequal Rational

1) $x^2 - 5x + 6 = 0$

$b^2 - 4ac$
 $(-5)^2 - (4 \times 1 \times 6)$
 $= 1$

6) $x^2 - 7x + 10 = 0$

$b^2 - 4ac$
 $49 - 40 = 9$

2) $2x^2 + 4x + 2 = 0$

$16 - 16$
 $= 0$

7) $4x^2 - 12x + 9 = 0$

$144 - 144 = 0$

3) $3x^2 - 2x + 5 = 0$

$4 - 60 \Rightarrow -56$

8) $2x^2 + 3x - 2 = 0$

$9 - (-16) \Rightarrow 25$

4) $x^2 + 6x + 9 = 0$

$36 - 36 \Rightarrow 0$

9) $x^2 - 4x + 8 = 0$

$16 - 22 \Rightarrow -6$

5) $5x^2 - 4x - 1 = 0$

$16 - (-20) \Rightarrow 36$

10) $3x^2 - 5x + 2 = 0$

$25 - 24 \Rightarrow 1$

Real and equal

imaginary

Real and equal

Real Unequal Rational

imaginary

Real and equal

Real Unequal and Rational

Real Unequal Rational

$$b^2 - 4ac$$

$$11.) x^2 - 4x + 2 = 0$$

$$16 - 8 = 8$$

$$> 0$$

$$12.) 3x^2 - 5x + 1 = 0$$

$$25 - 12 = 13$$

$$13.) x^2 - 3x + 1 = 0$$

$$9 - 4 = 5$$

$$\frac{3 \pm \sqrt{5}}{2}$$

$$\frac{3 + \sqrt{5}}{2}, \frac{3 - \sqrt{5}}{2}$$

$$\frac{4 \pm \sqrt{8}}{2} \Rightarrow \frac{4 \pm \sqrt{4 \times 2}}{2}$$

$$\Rightarrow \frac{4 \pm 2\sqrt{2}}{2}$$

$$\Rightarrow 2 \pm \sqrt{2}$$

$$2 + \sqrt{2}, 2 - \sqrt{2}$$

Real, unequal Roots
But Irrational
Roots

$$4 - \sqrt{3}$$

$$4 + \sqrt{3}$$

$$\frac{5 \pm \sqrt{13}}{2}$$

$$\rightarrow \frac{5 + \sqrt{13}}{2}, \frac{5 - \sqrt{13}}{2}$$

There is a quadratic equation, whose one roots $a + \sqrt{b}$

other $a - \sqrt{b}$

①

$$7 + \sqrt{5}$$

,

$$7 - \sqrt{5}$$

$$8 - \sqrt{7}$$

,

$$8 + \sqrt{7}$$